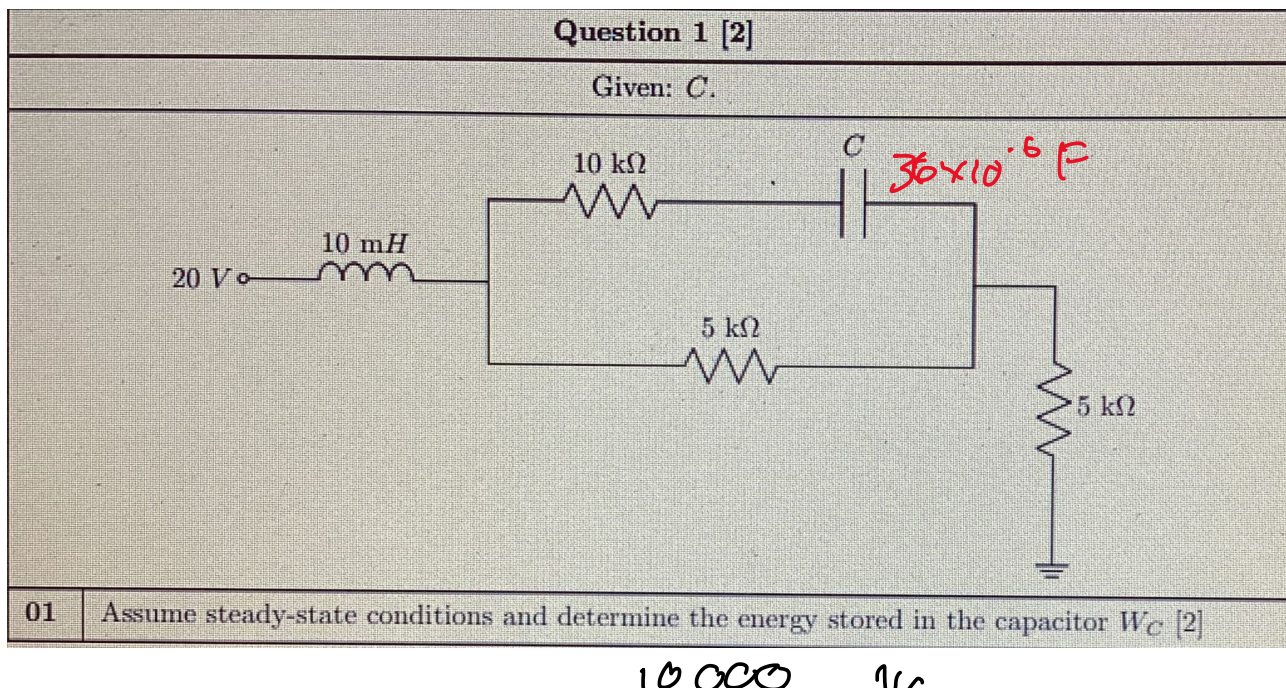




01 Assuming steady-state conditions and determine the energy stored in the capacitor W_C [2]

①

$W_C = \frac{1}{2} C V^2$

$= \frac{1}{2} (36 \times 10^{-6}) (10)^2$

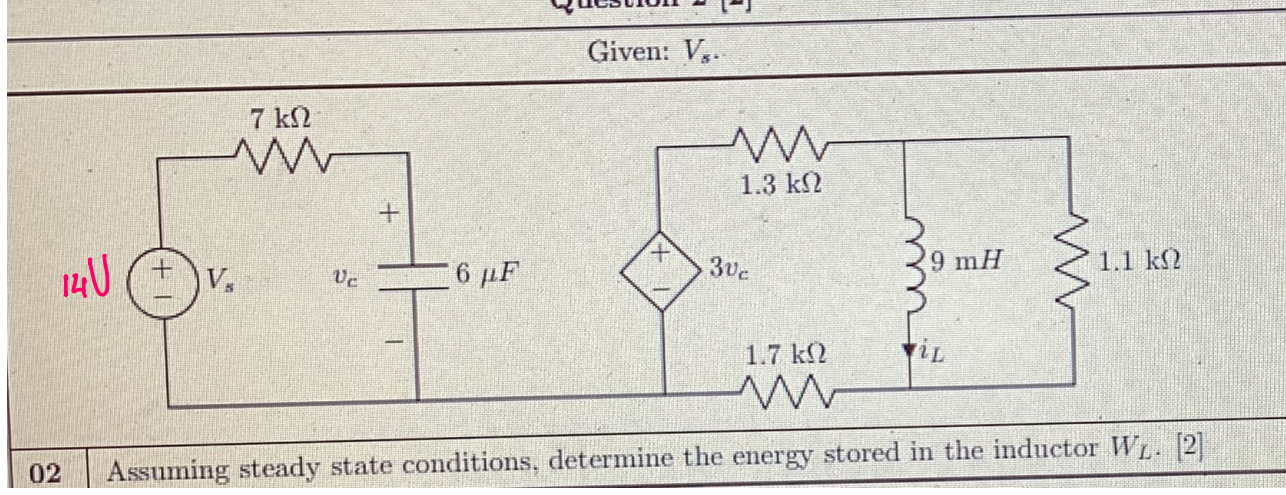
$= 1.8 \text{ mJ}$

$V_C = 5000 \times I$

$= 5000 \times \left(\frac{20}{10000} \right)$

$= 10 \text{ V}$

$I = \frac{20}{10000}$



02 Assuming steady-state conditions, determine the energy stored in the inductor W_L [2]

$W_L = \frac{1}{2} L i_L^2$

$= \frac{1}{2} (9 \times 10^{-3}) (0.014)^2$

$= 882 \text{ nJ}$

$V_C = 14 \text{ V}$

$\therefore$

$1300 I_L + 1700 I_L - 3(14) = 0$

$I_L = 0.014 \text{ A}$

Questions 3 to 6 [4]	
Given: X , Y , Z .	
Given the sinusoid $x(t) = X \cos(Yt + Z)$ V.	
03 Determine the angular frequency ω [1]	$\omega(t) = 77 \cos(125t + 100^\circ) \text{ V}$
04 Determine the amplitude A [1]	
05 Determine the phase angle θ [1]	
06 Determine the frequency f [1]	

③ $\omega = 12$

④ 77

⑤ 102°

⑥ $\omega = 2\pi f$

$f = \frac{\omega}{2\pi} = \frac{12}{2\pi} = 1.9099 \text{ Hz}$

Question 7 [2]	
Given: A , B .	
Two phasors are shown below:	
$V_1 = 8 \cos(\omega t - 45^\circ)$	$V_2 = 10 \cos(\omega t + 125^\circ)$
$I_1 = 10 \sin(\omega t - 87^\circ)$	$I_2 = 10 \sin(\omega t + 125^\circ)$
(3)	

07 By how many degrees is I_2 leading V_1 ? (Give your answer θ in the range from -180° to 180°)

⑦ $V_1 = 8 \cos(\omega t + 15^\circ)$

$I_2 = 10 \cos(\omega t + 125^\circ - 90^\circ)$

$= 10 \cos(\omega t + 32^\circ)$

$\therefore 8 \angle 15^\circ$

$10 \angle 32^\circ$

$\theta = 32^\circ - 15^\circ = 17^\circ$

Question 8 [2]	
Given: X .	
Given the sinusoid:	
$x(t) = -4 \cos(\omega t - 240^\circ) + 3 \sin(\omega t + 120^\circ) \text{ V}$	
Write $x(t)$ in polar form as $V = r \angle \theta$, with $r \geq 0$ and $-180^\circ \leq \theta < 180^\circ$.	
108 [V] []	

$-4 \cos(\omega t - 240^\circ) = 4 \sin(\omega t + 120^\circ)$

$= 4 \cos(\omega t - 240^\circ + 180^\circ)$

$= 4 \cos(\omega t - 60^\circ)$

$= 4 \angle -60^\circ$

$3 \sin(\omega t + 120^\circ) = 3 \cos(\omega t + 120^\circ - 90^\circ)$

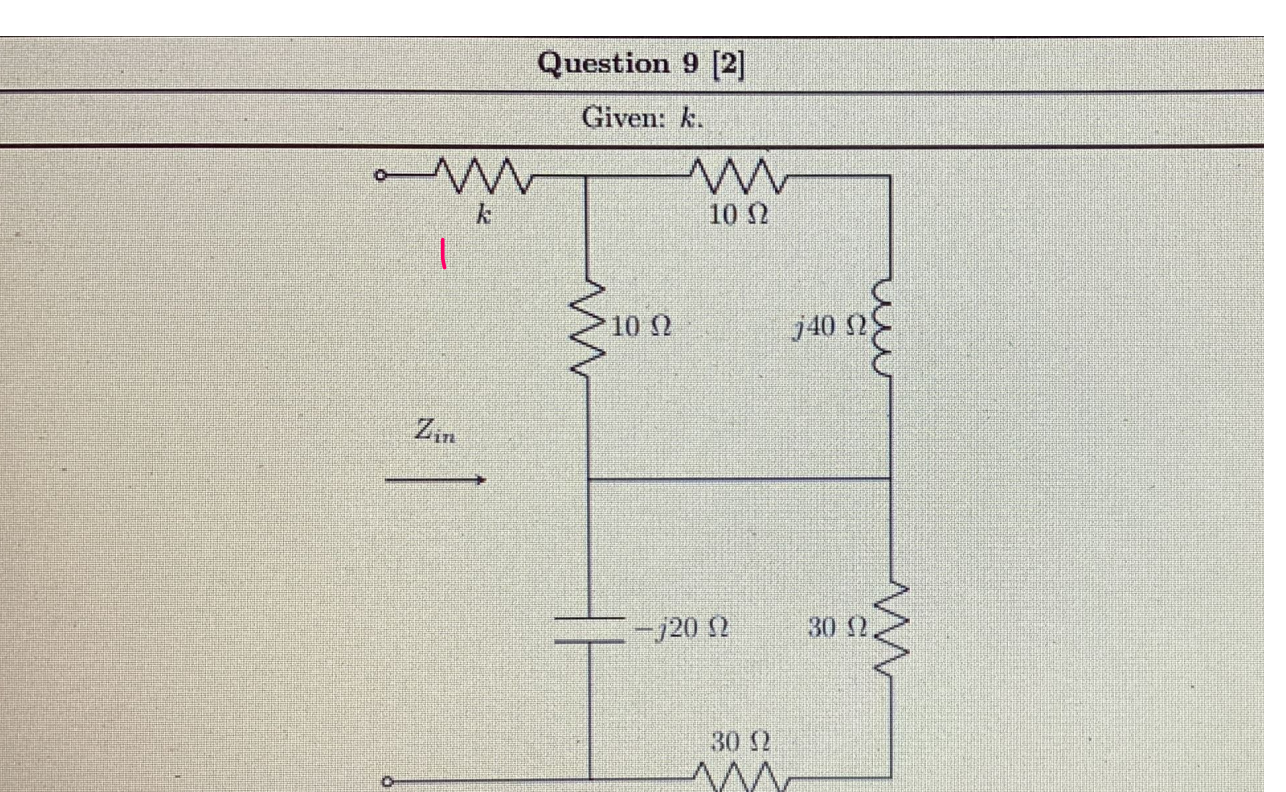
$= 3 \cos(\omega t + 30^\circ)$

$= 3 \angle 30^\circ$

$V(t) = 4 \angle -60^\circ + 3 \angle 30^\circ$

$= 4 \sqrt{5} \angle 3.4349^\circ$

$= 8.9443 \angle 3.4349^\circ$



09 Find Z_{in} in rectangular form [2]

$Z_{in} = 1 + R_{p1} + R_{p2}$

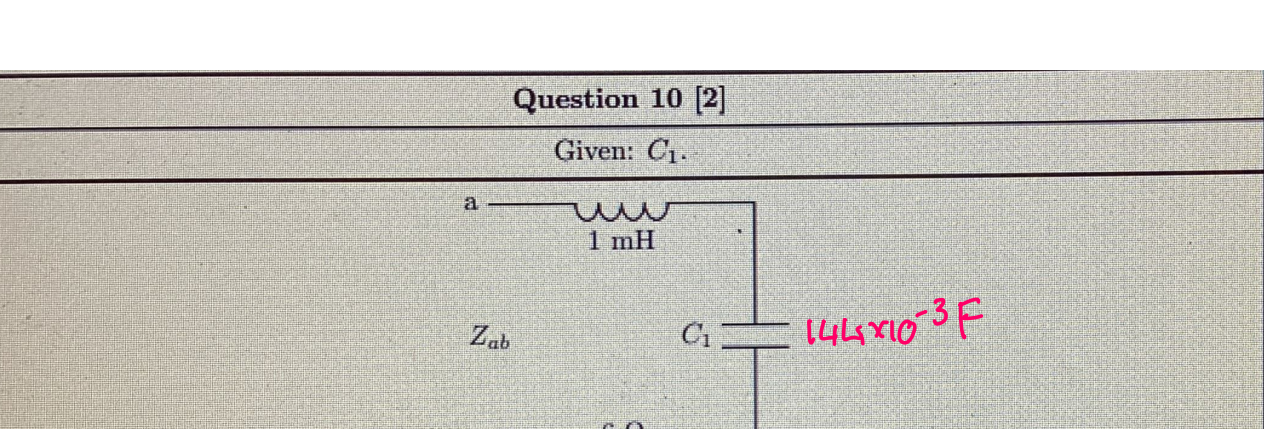
$= 16 - j16$

$R_{p1} = 9 + j2$

$R_{p2} = 6 - j18$

$\therefore Z_{in} = 1 + R_{p1} + R_{p2}$

$= 16 - j16$



10 At which angular frequency ω will the circuit below have a total impedance of $Z_{in} = 6 \Omega$ [2]

$Z_L = j\omega L = j\omega(1 \times 10^{-3}) \Omega$

$Z_C = \frac{1}{j\omega C} = \frac{1}{j\omega(100 \times 10^{-6})} \Omega$

$\therefore 6 = j\omega(1 \times 10^{-3}) + \frac{1}{j\omega(100 \times 10^{-6})} + 6$

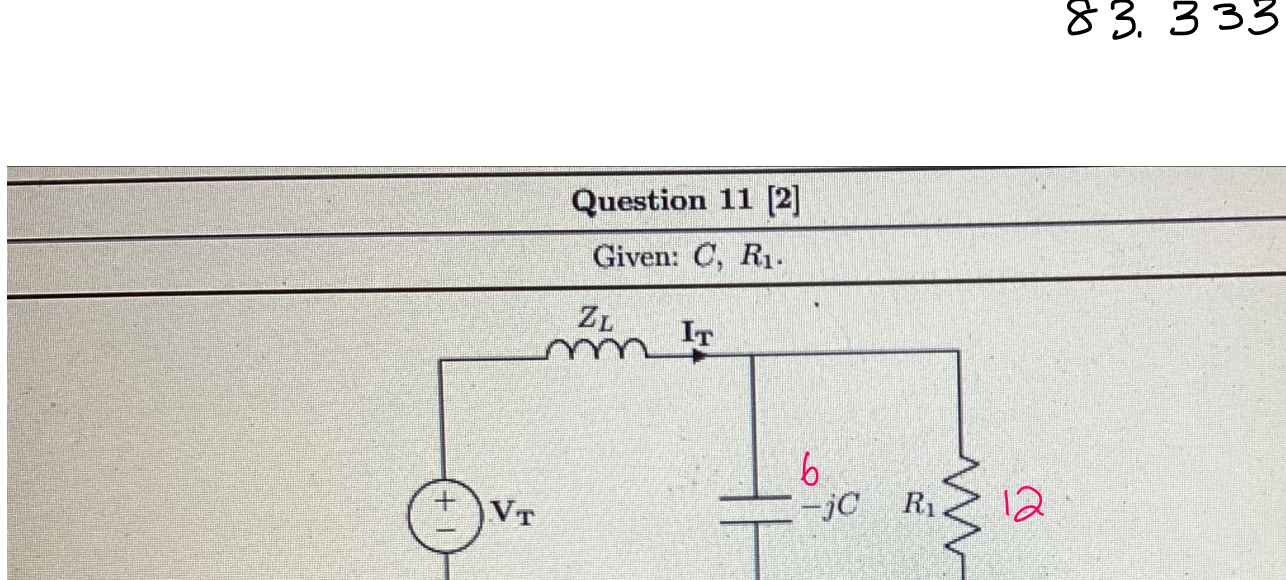
$(1 \times 10^{-3})j\omega = -\frac{1}{j\omega(100 \times 10^{-6})}$

$(1 \times 10^{-3})j\omega(100 \times 10^{-6})j\omega = -1$

$(-1)\omega^2 = -\frac{1}{(1 \times 10^{-3})(100 \times 10^{-6})}$

$\omega = \sqrt{\frac{1}{(1 \times 10^{-3})(100 \times 10^{-6})}}$

$= 83.3333333$



11 Give the value of Z_L so that the phases of V_T and I_T differ by 0° [2]

$V_T - I_T = 0 \Rightarrow \frac{V_T}{I_T} = 1$

$V_T = I_T (24 - j4.8)$

$(\frac{1}{12} + \frac{1}{-j6})^{-1} = 24 - j4.8$

$V_1 = (\frac{6}{6+4}) \times 40$

$V_2 = (\frac{4}{6+4}) \times 40$

Voltage division:

$V_1 = (\frac{24 - j4.8}{(24 - j4.8) + Z_L}) V_T$ (1)

Ohm's law:

$V_1 = I_T (24 - j4.8)$ (2)

Sub (2) in (1)

$I_T (24 - j4.8) = (\frac{24 - j4.8}{(24 - j4.8) + Z_L}) V_T$

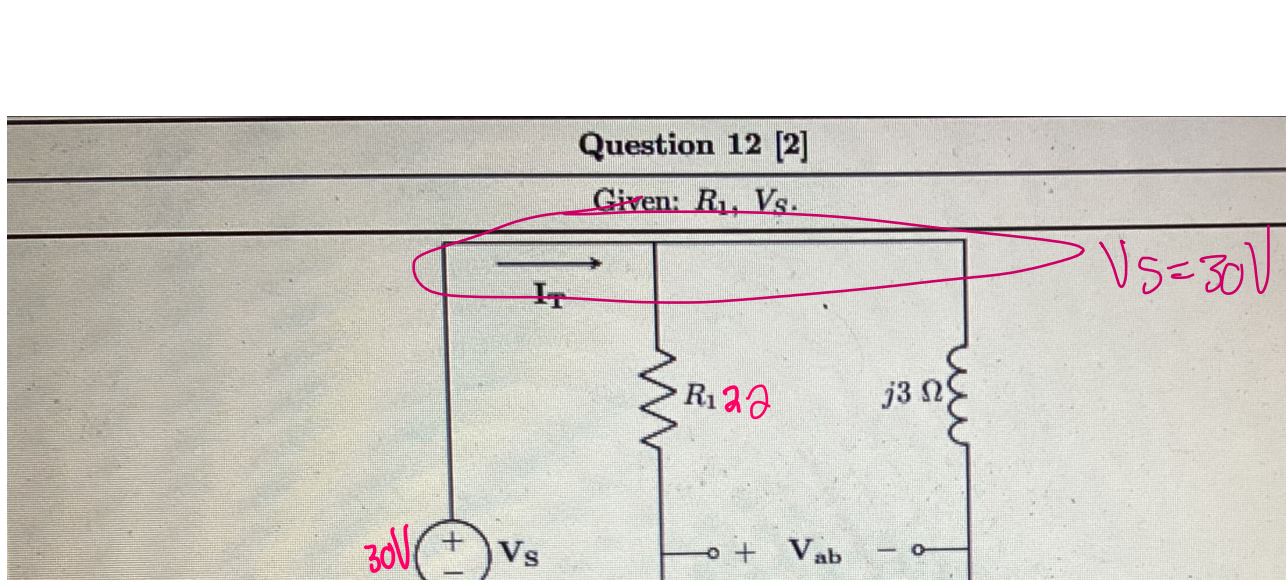
$(24 - j4.8) = (\frac{24 - j4.8}{(24 - j4.8) + Z_L}) \frac{V_T}{I_T} = 1$

$(24 - j4.8)(24 - j4.8) + Z_L(24 - j4.8) = 24 - j4.8$

$Z_L = \frac{(24 - j4.8) - (24 - j4.8)}{24 - j4.8}$

$Z_L = -1.4 + j4.8$

$\therefore j4.800$



12 Determine I_T in rectangular form [2]

$\frac{30}{22 - j15} + \frac{30}{4 + j3} - I_T = 0$

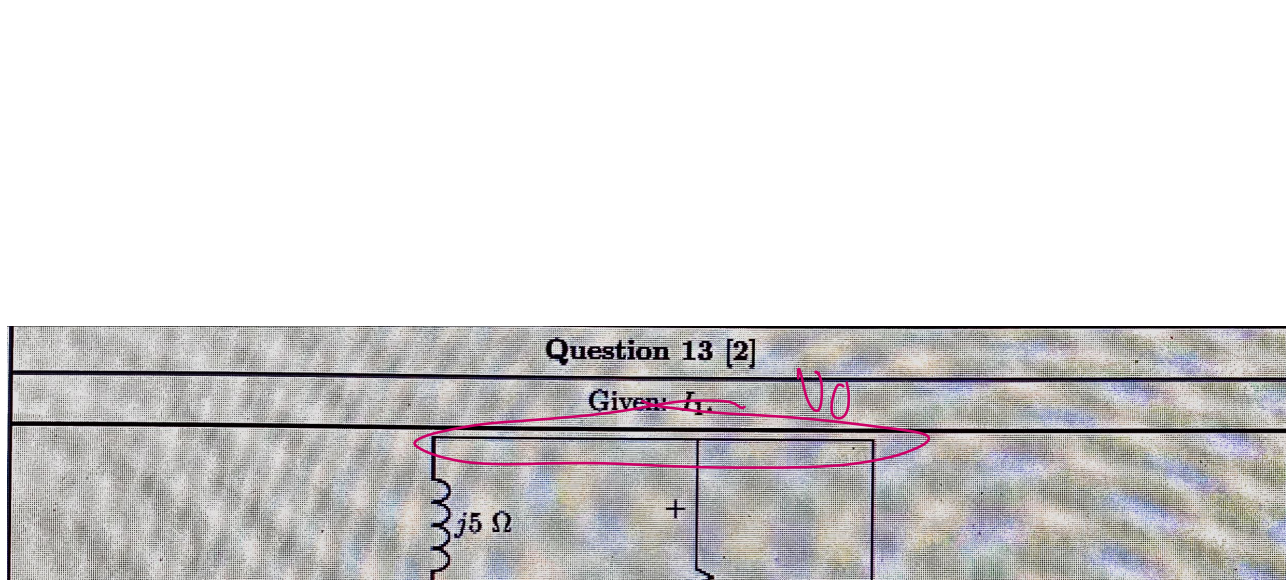
$I_T = \frac{30}{22 - j15} + \frac{30}{4 + j3}$

$= 5.7309 - j2.9653$

$\frac{V_a - 30}{3} + \frac{V_a}{-j15} = 0$ (1)

$\frac{V_b - 30}{-j3} + \frac{V_b}{4} = 0$ (2)

$V_a - V_b = V_{ab}$



13 Determine V_0 in polar form [2]

$\frac{V_0}{5} + \frac{V_0 - 2.5 I_D}{j5} - 15 = 0$

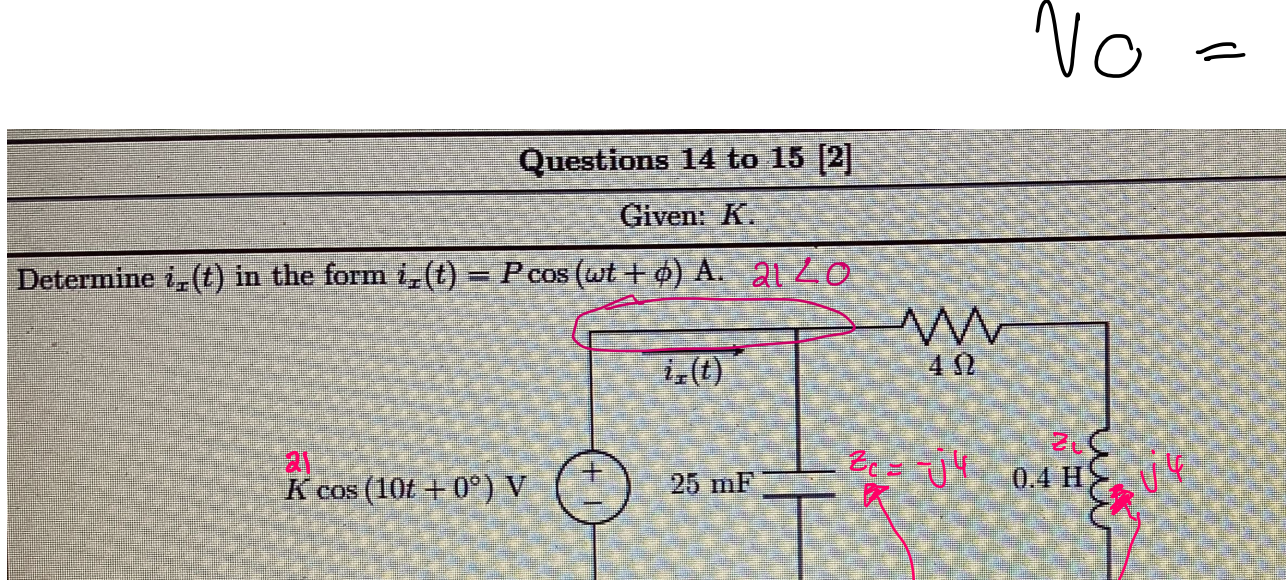
$\frac{V_0}{5} + \frac{V_0 - 2.5 (\frac{V_0}{5})}{j5} - 15 = 0$

$\frac{1}{5} V_0 + \frac{1}{j5} V_0 - \frac{0.5}{j5} V_0 = 15$

$(\frac{1}{5} - j\frac{1}{5}) V_0 = 15$

$V_0 = 60 \angle 30^\circ$

$I_D = \frac{V_0}{5}$



14 Provide P where $P > 0$ [1]

15 Give θ where $-180^\circ \leq \theta < 180^\circ$ [1]

$Z_C = \frac{1}{j\omega(25 \times 10^{-3})} = -j4$

$Z_L = j\omega(0.4) = j4$

$A < 0$

$\frac{2 \angle 0}{-j4} + \frac{2 \angle 0}{4 + j4} - I_X(1) = 0$

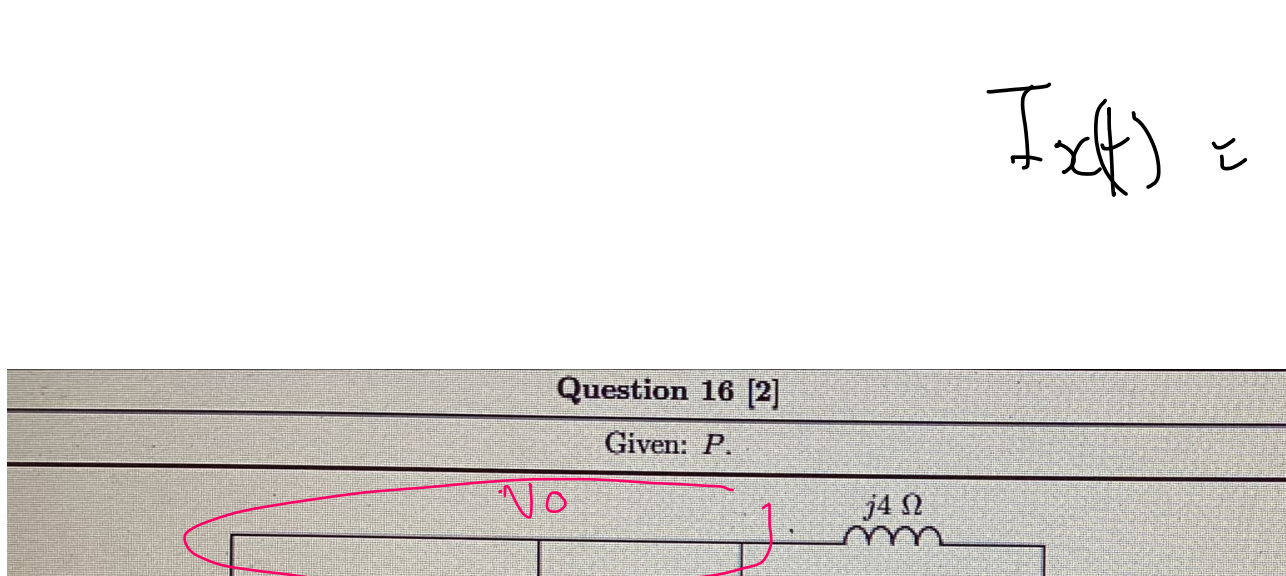
$I_X(1) = \frac{2}{8} + j\frac{2}{8}$

$= 3.7123 \angle 45^\circ$

$I_X(t) = 3.7123 \cos(10t + 45^\circ)$

$\therefore P = 3.7123 \text{ A}$

$\Phi = 45^\circ$



16 Determine V_0 in rectangular form [2]

$\frac{V_0}{10} - (60 \angle -90^\circ) + \frac{V_0}{-j5} + \frac{V_0 - (8 \angle 0)}{j4} = 0$

$(\frac{1}{10} - j\frac{1}{20}) V_0 = (60 \angle -90^\circ) + (\frac{8 \angle 0}{j4})$

$V_0 = 248 - j496$